

Day 1 Presentation

Robert Gulotty

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R + RStudio

- ▶ For this class, we will use R, but interact with it using RStudio.
 - ▶ R is a flexible language for data analysis.
 - ▶ RStudio is an IDE, or ***Integrated Development Environment***, which offers an interface to R.
- ▶ Your assignments can all be turned in pdf format using Latex + Rmarkdown, which is built into RStudio.

Today's lesson plan

Today we will cover

- ▶ Basic RStudio interface.
- ▶ Workflow and related issues.
- ▶ How to get help and ask programming questions.

Main lesson from Hadley Wickham



CRAFTSMANSHIP

“You're doing it right if you get frustrated: if you're not frustrated, you're (probably) not stretching yourself mentally”

- Hadley Wickham

Figure 1: Motivational Quote

Principles to keep in mind

- ▶ You can learn this.
- ▶ It will take time and practice.
- ▶ You will not break anything important.

Rstudio Panes

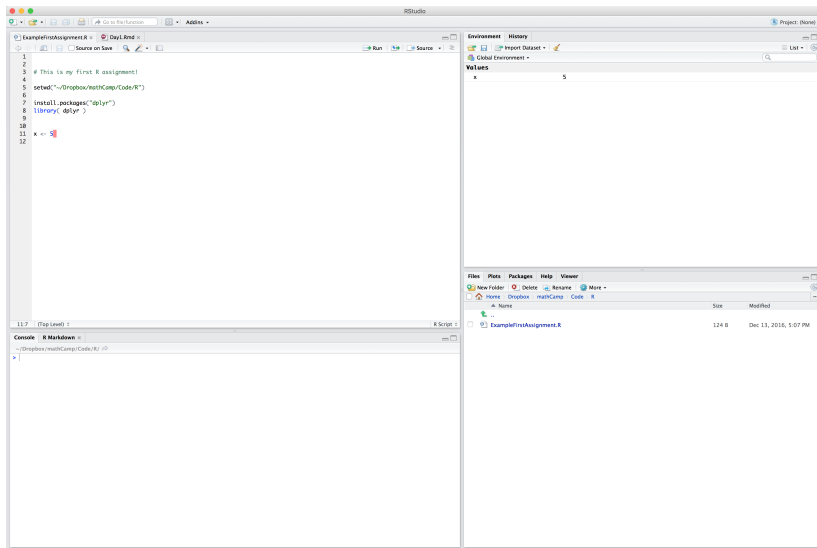


Figure 2: RStudio In Action

Source: Where your code is edited.

The top left part of the screen is the Source.

- ▶ This is where you type in your code, in what is called a script.
- ▶ The default document is a .R document.
- ▶ The more advanced documents are called Rmarkdown documents, .Rmd
- ▶ RStudio highlights syntax and autocompletes code (tab)

Console: Where your code is run.

The bottom left part of the screen is the Console.

- ▶ This is an interactive code environment.
- ▶ The little `>` is where the code is input.
- ▶ A `+` means that it is waiting for a command. `esc` to cancel.
- ▶ `control - l` to clear the Console.

Environment

The top right part of the screen has the Environment and History tabs.

- ▶ Environment includes all the objects in RAM.
- ▶ As you load or create data, this populates your Environment
- ▶ The little grid icon to the right of Data objects opens a “view” screen.
- ▶ `ls()` returns all the objects in the global environment.
- ▶ `rm(x)` removes an object `x`.

Objects and Functions in R

- ▶ There are several fundamental datatypes, which you will learn.
 - ▶ Characters: strings enclosed in "Colorado"
 - ▶ Numeric: integers and doubles 21.1, 4L, 9e2
 - ▶ Logical: TRUE, FALSE, NA
- ▶ Functions have a name, followed by ()

```
rnorm(10)
```

```
## [1]  1.1129723 -1.5713803 -1.2295564  
## [4] -1.0248712 -0.2903676  1.2901336  
## [7]  0.5642119 -0.2249501  0.3737685  
## [10] -0.4495167
```

Useful Shortcuts

Purpose	Keyboard shortcut	Menu Path
Run line	command - enter	Code / Run Selected Line(s)
Code Completion	tab	
Clear Console	control - l	Edit / clear console
Save Script	command - s	File / save
Reformat Script	command - a	Code / Reformat Code

These are all available in the Help Menu

Try your first R language Commands

- ▶ Type `x <- 5^6` into a script, put your cursor on that line, press command-enter.
- ▶ Try the same thing, but put a `#` before it
- ▶ Help: `?plot`

Workflow

Set-up for assignments in Rmarkdown

- ▶ R markdown documents are created in the File menu and appears in the Console.
- ▶ We have provided templates, with lots of code examples.
- ▶ R markdown has embedded **r code** and embedded **latex** code.

LaTeX Code

- ▶ LaTeX was invented to typeset mathematics:

`$\int_{i=1}^{\infty} \frac{1}{x} = \delta$`

gives you

$$\int_{i=1}^{\infty} \frac{1}{x} = \delta$$

- ▶ There is a template sheet available on Chalk with many useful commands.

How to get Help

Help in R

- ▶ `? + the command`. eg. `?plot`, `?'=='`, `?'%in%'` loads help for **loaded** functions.
- ▶ `help( package="dplyr")` gives the main page for a package.
- ▶ `?? + command` searches R help pages.

How to google it.

- ▶ Add CRAN + google search term
- ▶ Add [r] + google search term
- ▶ Go to datacamp.com, [stackoverflow](https://stackoverflow.com)

How ask for help online.

- ▶ Isolate one issue. Each issue gets it own minimal complete verifiable example.
- ▶ Try to make the mistake happen in as few lines of code as possible (minimal/verifiable).
- ▶ Then send the whole script (complete).

Example

```
# Goal: find the easiest language to learn

# Fake data:
df1 <- data.frame(codetype = c("R", "Matlab", "Stan"),
                  learningtime = c(20, 15, 30))

# The offending code snippet:
df1[which.min(df1), ]

# Description of error message:
# Error is "(list) object cannot be coerced
# to type double"
```